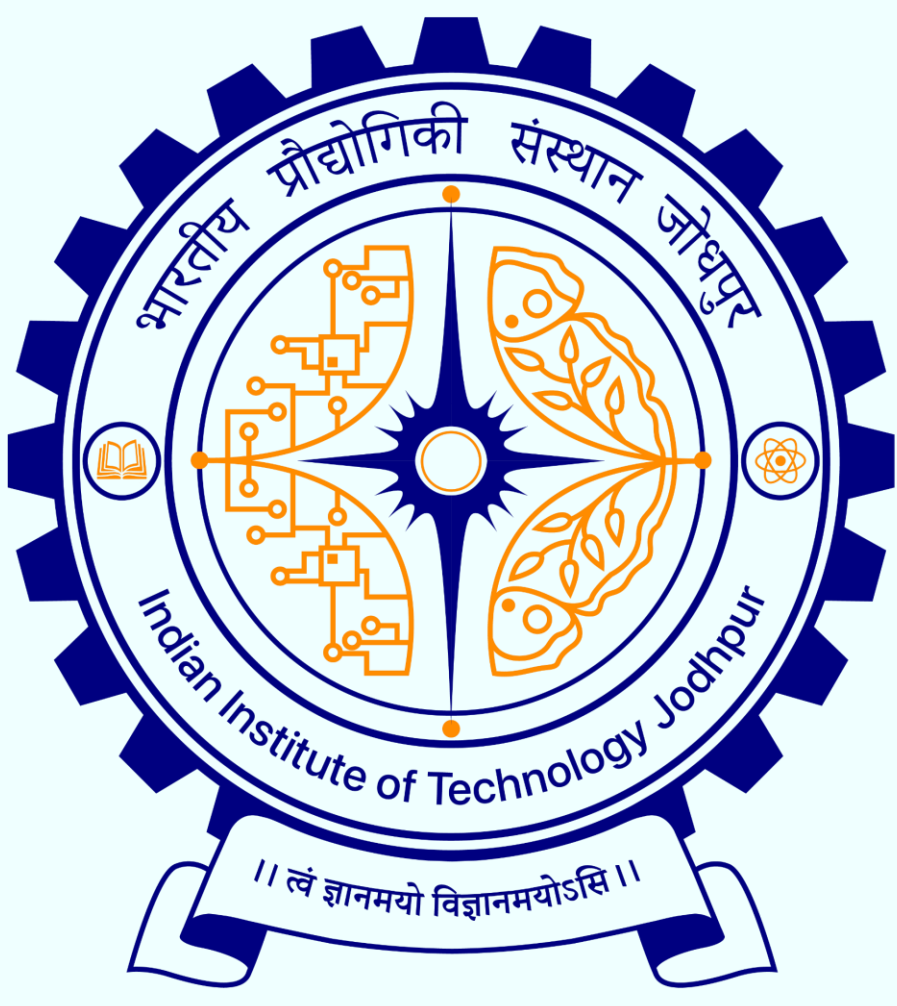




Second-order Correlation for the Classical and Quantum Light Sources using Timepix Camera

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INTRODUCTION

Precise measurement of photon arrival times is crucial for research in physics, chemistry, and the life sciences. The Hanbury Brown and Twiss (HBT) interferometer serves as a fundamental tool for photon correlation studies, enabling exploration of light coherence, statistical behavior, and quantum properties. Coincidence detection across multiple detectors provides insights into non-local correlations and entanglement, while similar methods in nuclear physics help reconstruct decay pathways and probe fundamental interactions.

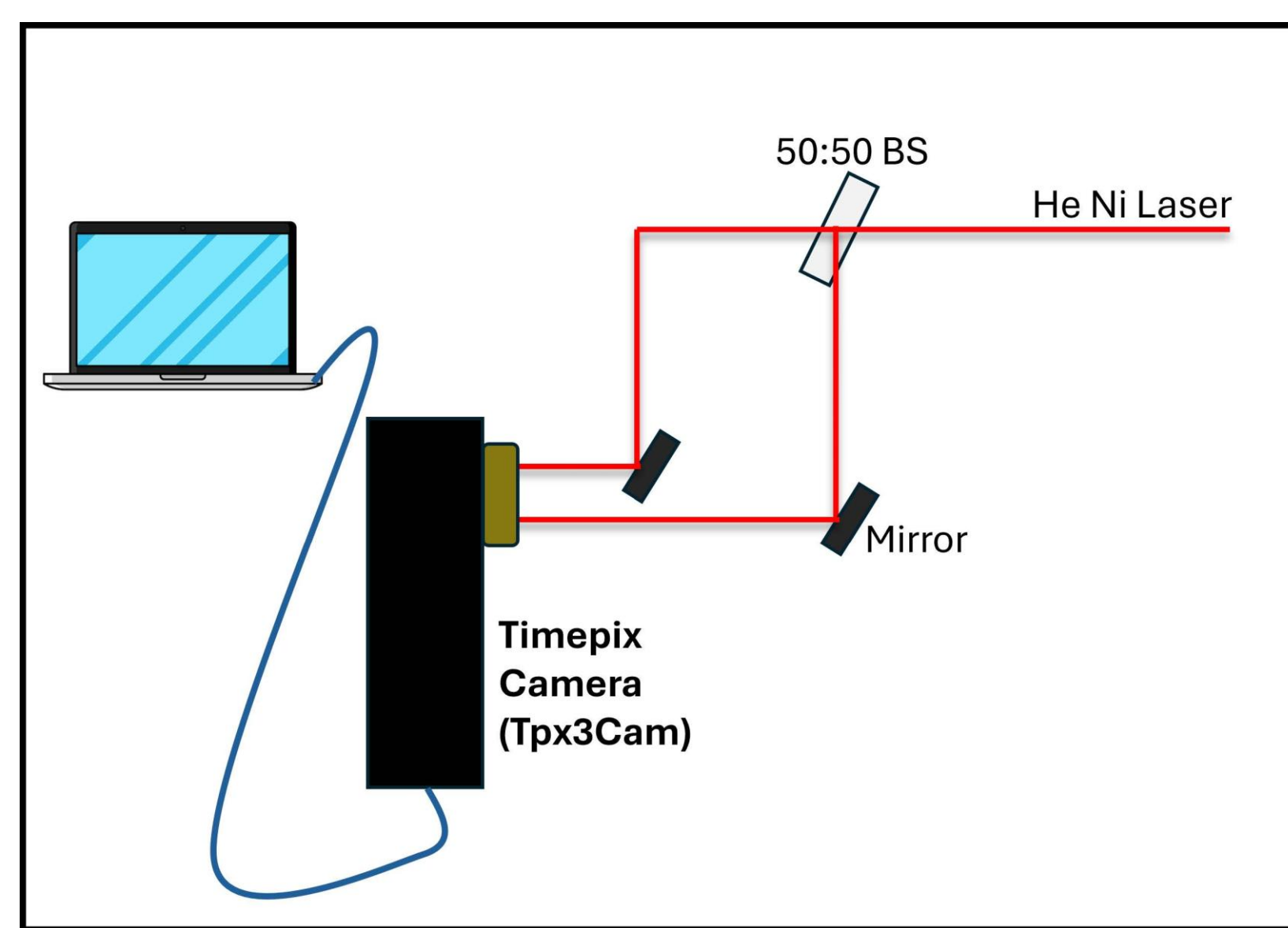
In this work, we utilize the Tpx3Cam (AMS), a high-resolution, event-based imaging camera, to study second-order photon correlations of a coherent light source and benchmark its performance against a commercially available Time Tagger (Swabian Time Tagger 20).

TIMEPIX CAMERA (TPX3CAM)

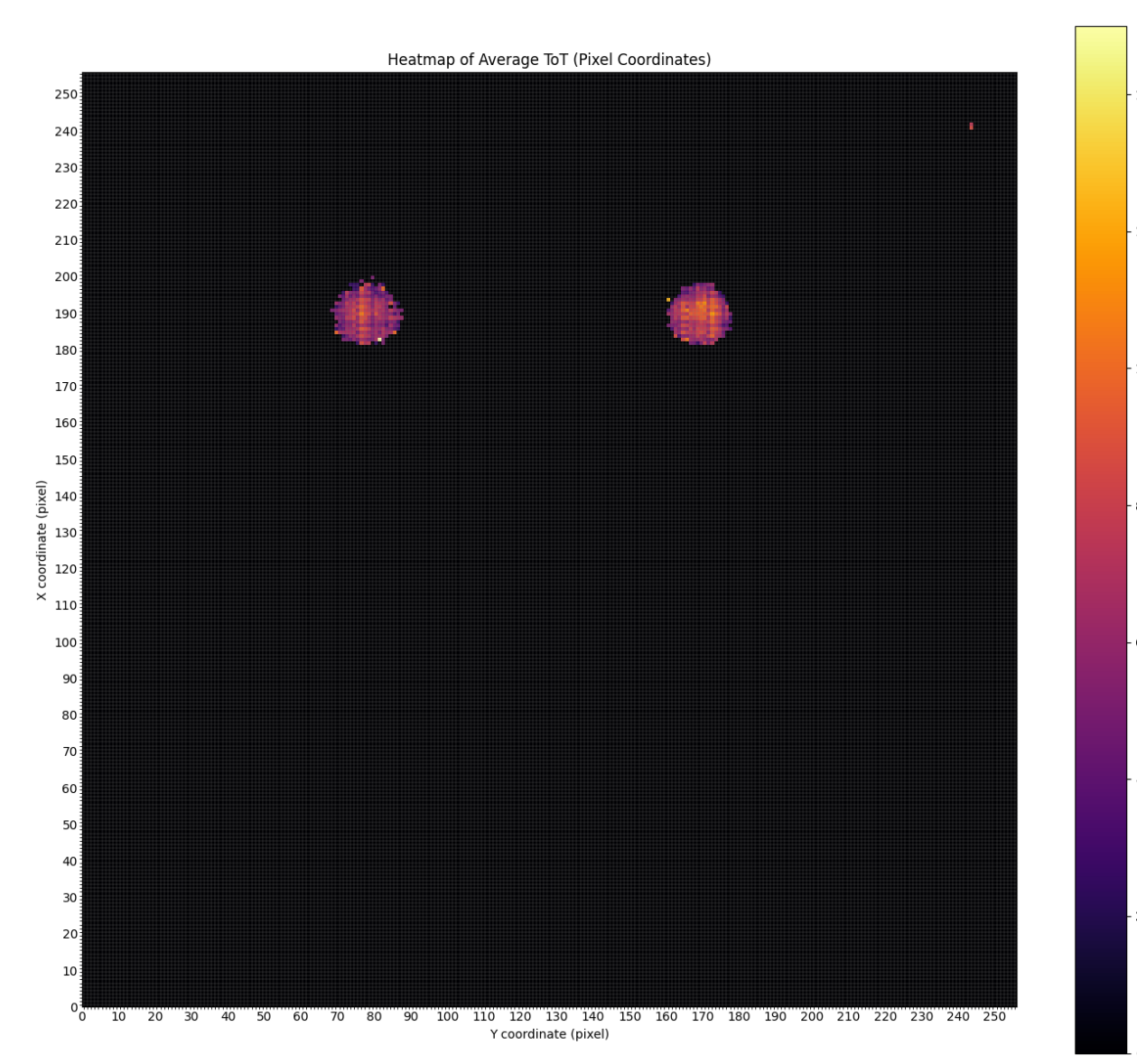
- Single-photon sensitivity with 1.6 ns temporal resolution using an image intensifier.
- 256 × 256-pixel array (55 × 55 μm each), with each pixel operating independently.
- Records both Time of Arrival (TOA) and Time over Threshold (TOT) for photon timing and flux.
- Event-based, data-driven readout — only active pixels are recorded (no global trigger needed).
- Supports throughput up to 80 MP/s and external trigger inputs.
- Enables fast, precise photon timing for quantum optics and correlation studies.



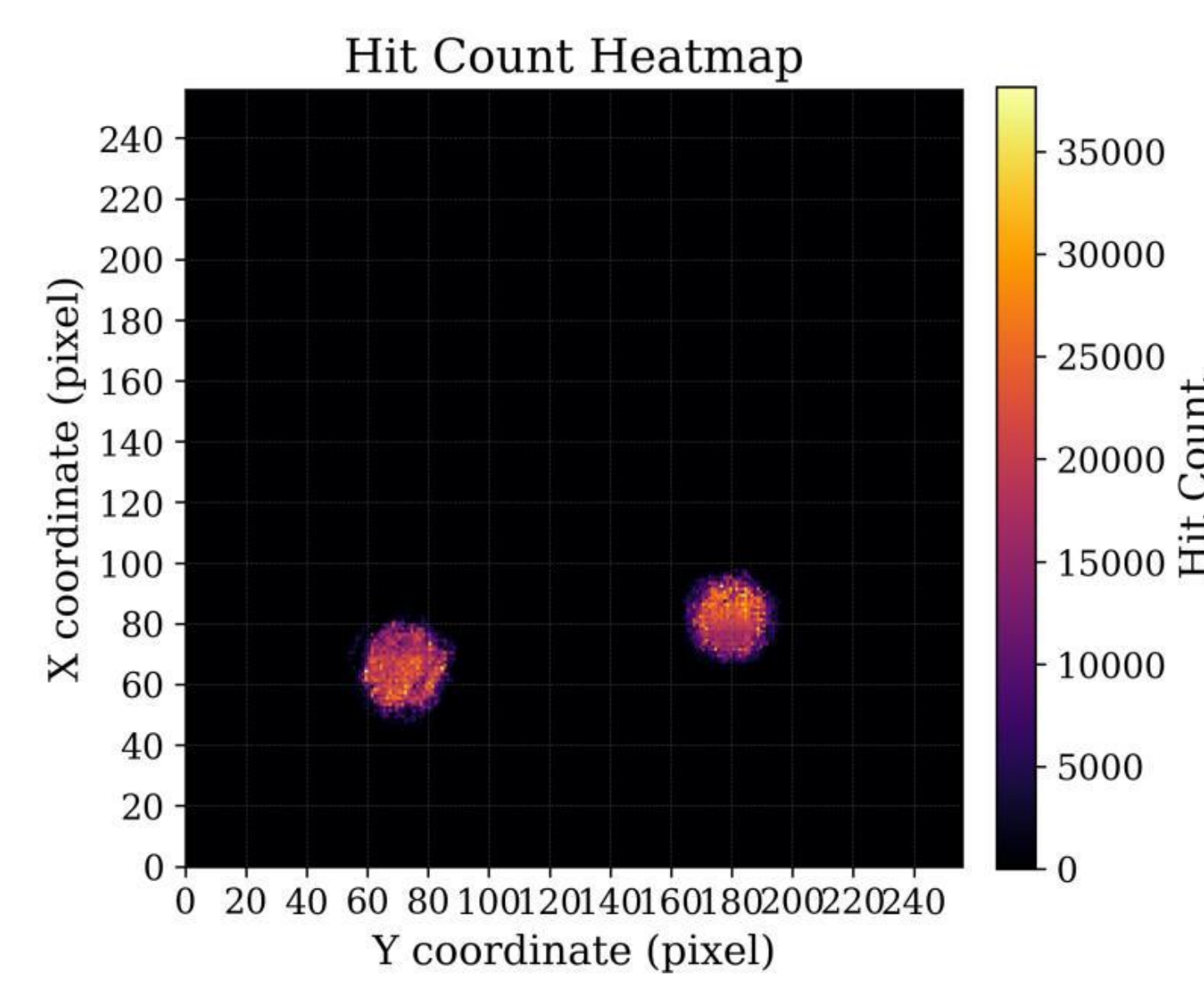
METHODOLOGY



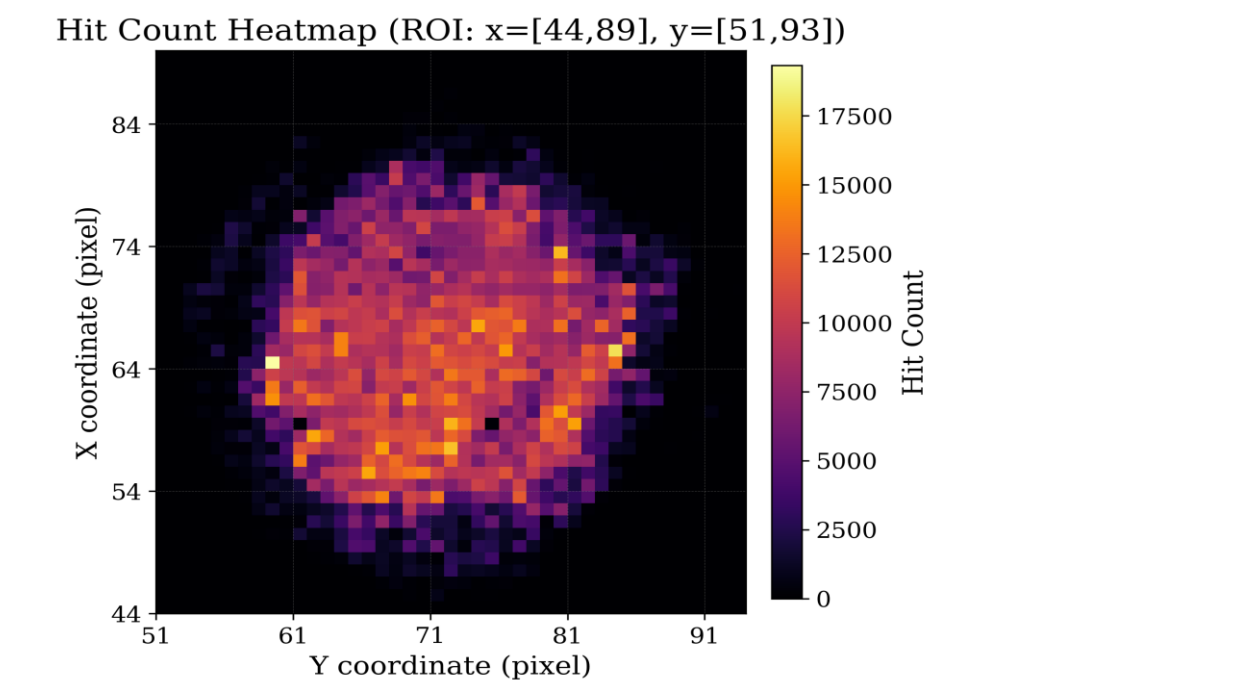
(a) Experimental Setup



(b) Unfiltered Data



(c) Filtered Data



(d) Channel 1 Enlarged

	A	B	C	D	E	F	G	H	I	J
ChipNum	0	0	0	0	0	0	0	0	0	0
Mode	0	0	0	0	0	0	0	0	0	0
TriggerCounter	0	0	0	0	0	0	0	0	0	0
TDC_timestamp	0	0	0	0	0	0	0	0	0	0
GlobalTime	0.0011387	319500	25	77	179					
ChipToA	0.001139413	320200	125	82	188					
Tot	0.001139488	320275	125	84	191					
x	0.001139489	320275	125	75	187					
y	0.001139239	320025	200	81	181					
x	0.001139313	320100	300	86	185					
y	0.001139425	320225	100	78	187					
x	0.001139441	320225	100	89	186					
y	0.00113937	320150	90	94	185					
x	0.001139381	320175	125	77	174					
y	0.001139478	320275	125	72	187					
x	0.001139502	320300	100	91	185					

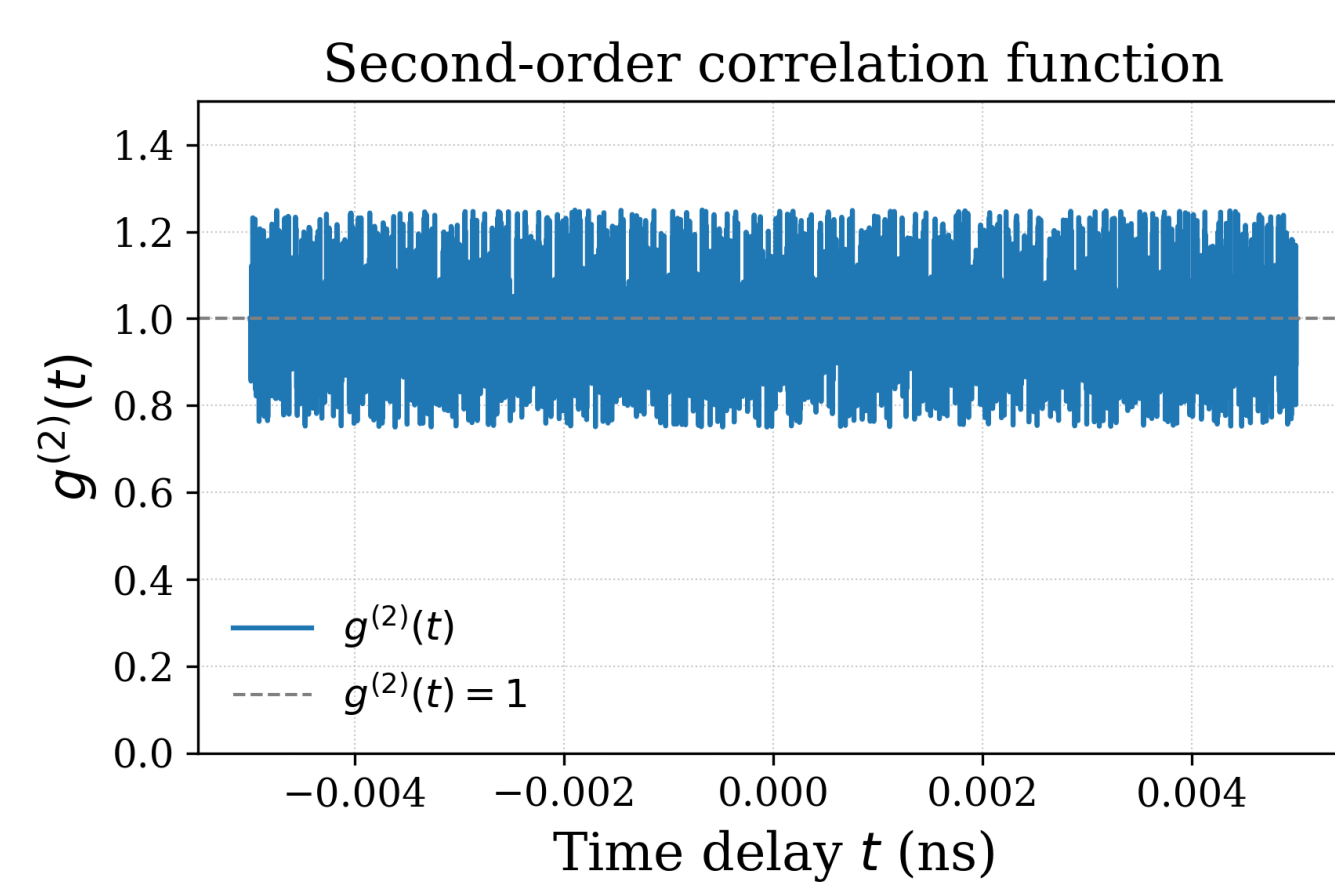
(e) Data



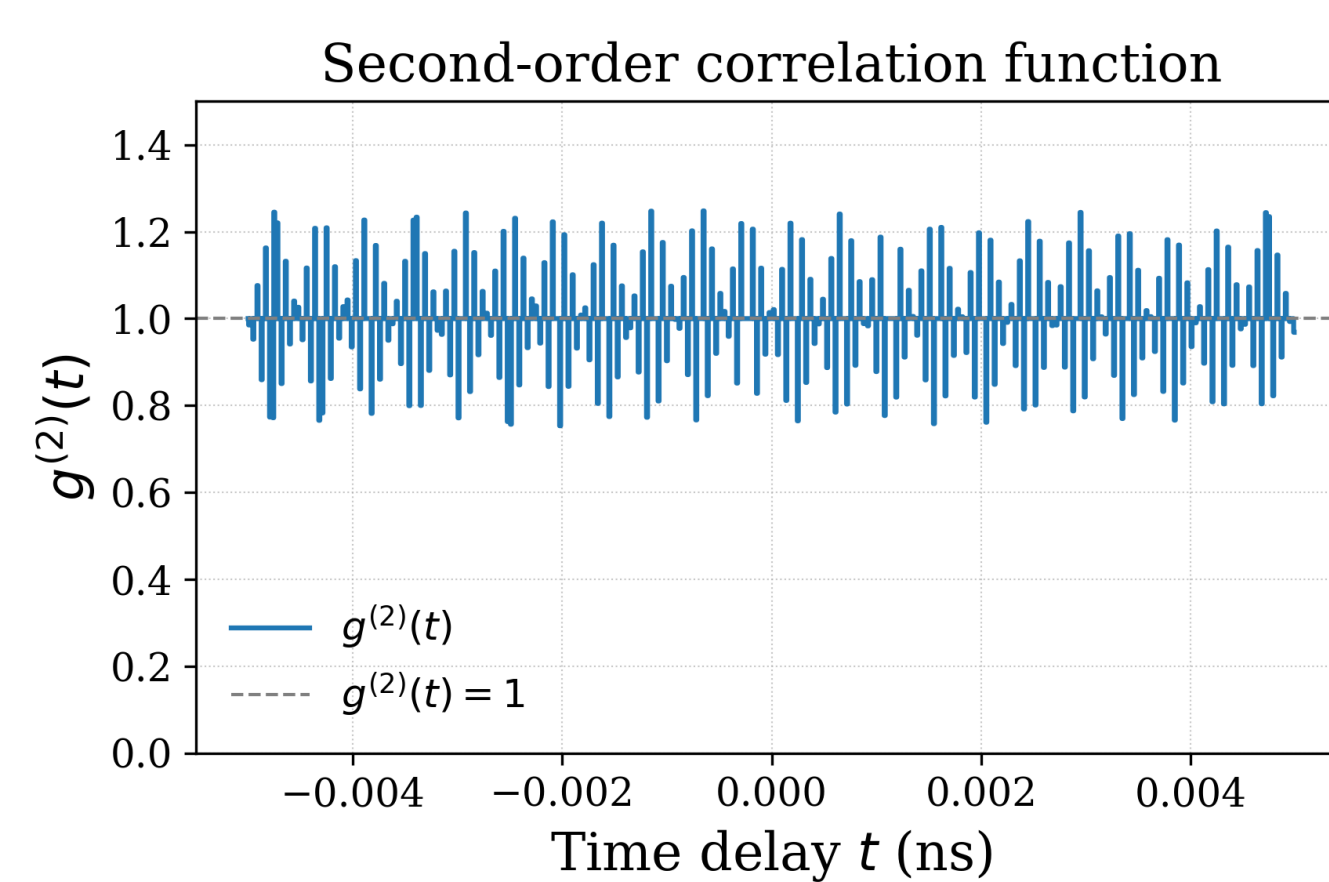
The histogram was then normalized to obtain the second-order correlation function:

$$g^2(\tau) = \frac{\Delta t}{\text{Binwidth}(\tau)N_1N_2} \text{Histogram}(\tau)$$

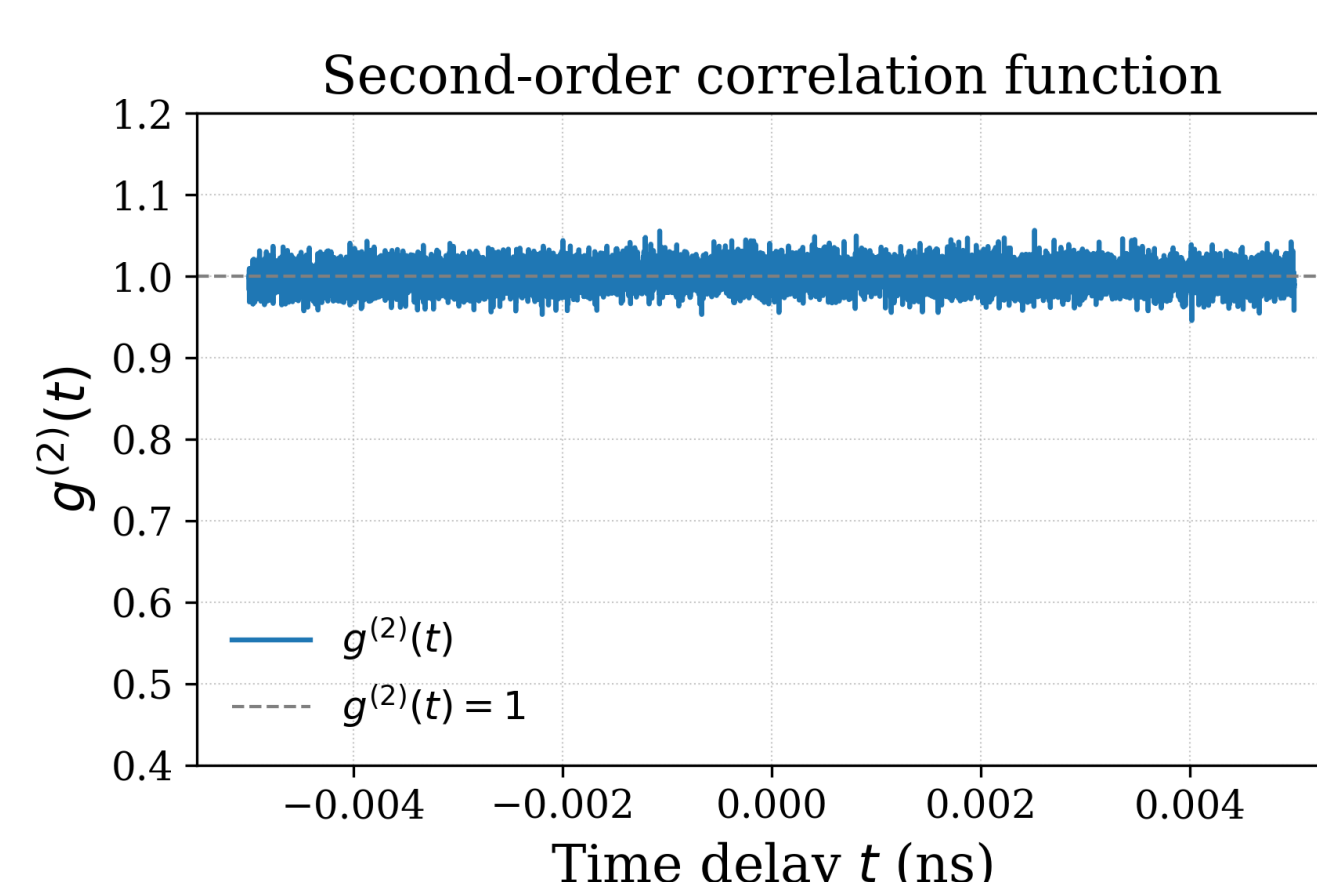
RESULTS AND DISCUSSION



(a) Tpx3Cam - 1 μs binwidth



(b) Tpx3Cam - 10 μs binwidth



(c) Swabian Time Tagger 20 - 1 μs binwidth

- The second-order intensity correlation, $g^2(\tau)$, of a He-Ne laser was measured using Tpx3Cam.
- As expected for a coherent light source, the correlation function was consistently unity ($g^2(\tau)=1$) across all time delays.
- These results showed excellent agreement with data obtained from a Swabian Time Tagger 20.

CONCLUSION

We employed the Tpx3Cam for the investigation of second-order intensity-intensity correlation $g^2(\tau)$ for the coherent light source. We got the expected unity values for all the delays and compared them with the Swabian Time Tagger 20. In the future, we will investigate it for all the light sources, such as thermal and quantum light sources.

ACKNOWLEDGEMENT

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